IV Year B.Tech II Semester– Major Project Review

APPLICATION FOR ISSUING DIGITALLY SIGNED ACADEMIC CERTIFICATES

Team Members Details

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IV Year B. Tech II Semester – CSE (Cyber Security)

Under the Guidance of

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Abstract

- With increased emphasis on Digital Technology by Govt., Non-Govt. agencies at various levels, a need is felt to use all the available technologies towards going Digital which has many advantages such as Efficiency, Traceability, Confidentiality, Integrity of documents and Origin, Non-repudiation etc. It also has the benefit of being environment friendly.
- The System acts as a tool for the Controller Of Examinations (COE) to generate bulk digitally signed academic documents by providing student academic data input and a digital certificate and key store(a PKI physical token). The system is developed using Python and Java and their supported text/image/crypto libraries.
- Through its user-friendly Graphical User Interface (GUI), the system collects details of the student academic performance as .xls file, the scanned signature , the photo etc. and outputs the Digitally Signed certificates with all the fields properly laid out.

Introduction

- The Digital India movement emphasizes the need to transition existing systems to a paperless environment. When dealing with critical data such as academic certificates, ensuring security and authenticity is paramount. Cryptography, the science of securing data through encryption, helps protect sensitive information from unauthorized access and manipulation.
- Cryptography plays a key role in securing data by encrypting information to prevent unauthorized access. A digital signature is a specific type of encryption that protects an individual's signature from forgery. It verifies the identity of the sender and ensures message integrity
- A digital signature works by encrypting the hash value of a document using the sender's private key, ensuring that the signature is unique to each document. Since the sender’s public key is widely available, anyone can verify the document’s authenticity and confirm it hasn't been altered. This technology is widely used in secure communications, financial transactions, and academic certifications.

Existing System

- **Manual Digital Signature Systems**: Certificates are manually signed using software like Adobe Acrobat or Microsoft Word, requiring individual processing for each document. This approach is time-consuming, prone to errors, and unsuitable for bulk processing.
- **Semi-Automated Systems**: These systems use scripting languages like Python and Java to automate certificate generation by importing data from spreadsheets (.xls, .csv). While they offer partial automation, they lack scalability, user-friendly interfaces, and require technical expertise.
- **Web-Based Certificate Issuance Portals**: Platforms such as National Academic Depository (NAD) and DigiLocker provide online access to certificates, offering secure storage and verification through APIs. However, they require internet connectivity, involve potential security vulnerabilities, and have high implementation costs.

Proposed System

- The system allows the authorized sender (e.g., institution administrator) to log in securely using a token password and a key password to access the platform.
- The authorized sender selects multiple certificates for processing in bulk.
- The platform applies cryptographic techniques and hash functions to generate a unique digital signature for each certificate
- The digital signature is applied to the certificates using the private key of the sender, ensuring the integrity and authenticity of the documents.
- The system supports printing multiple certificates in bulk after applying the digital signature.

- All signed certificates are logged securely in the system to maintain a record for future reference.
- The platform ensures restricted access to sensitive keys, passwords, and signing operations through encryption and user authentication.
- Notifications or reports will indicate the successful completion of the signing and printing process.
- In the proposed system, an additional layer of security is implemented by requiring users to authenticate themselves before downloading their digitally signed academic certificates. Similar to the Aadhaar download process, users will need to enter the first four letters of their name along with their year of birth to verify their identity and access their certificate securely.

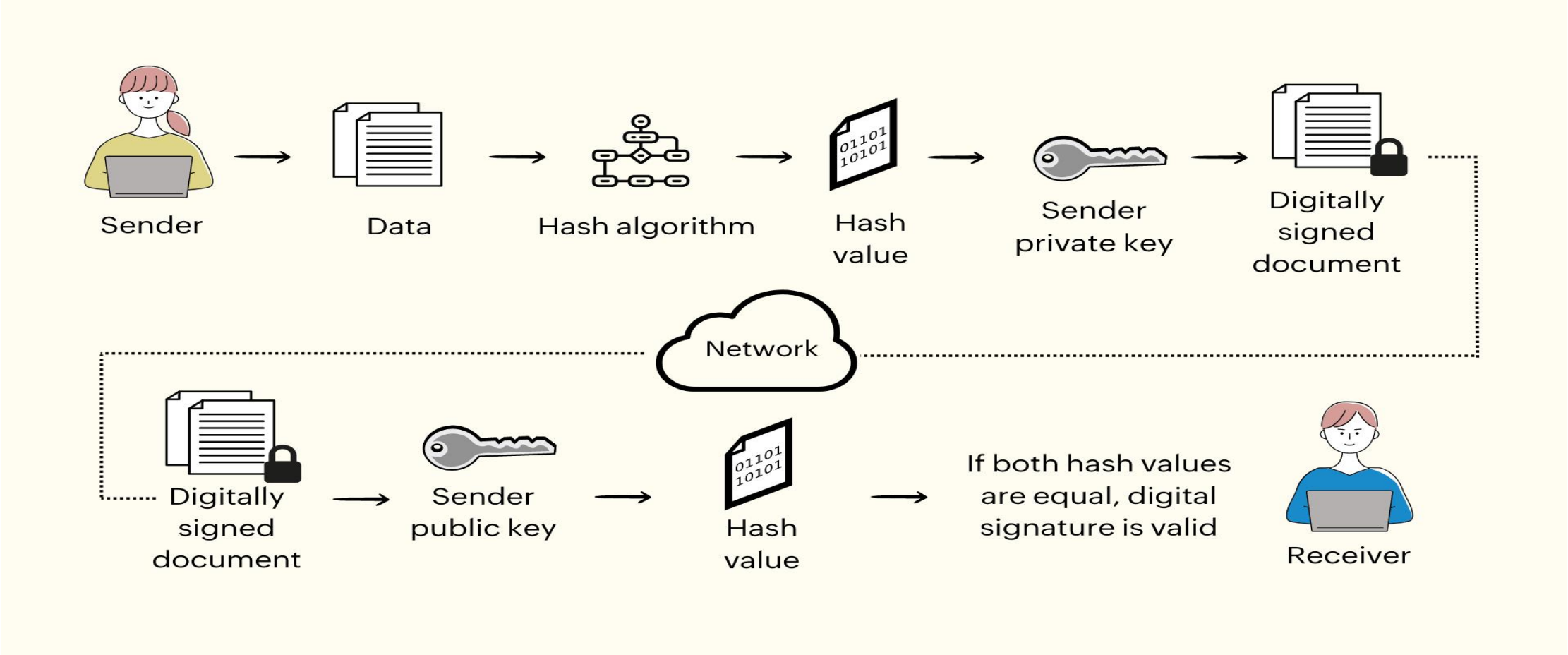
Existing vs Proposed System

Feature	Existing System	Proposed System
Certificate Issuance	Manual process requiring physical signatures and stamps	Digital signing using cryptography and hash functions for enhanced security.
Authentication	No strong authentication for certificate issuance.	Uses a token password and key password to authenticate authorized users.
Security	Certificates can be easily forged or tampered with..	Secure digital signatures ensure authenticity and prevent forgery
Batch Processing	Certificates are issued and printed one by one.	Supports bulk processing to sign and print multiple certificates at a time.
Verification	Requires manual verification by contacting the institution.	Digital signatures allow instant verification without contacting the issuer.
Efficiency	Time-consuming and error-prone manual work.	Automated issuance reduces errors and speeds up the process.

Literature Review

S. No	Paper Title, Authors & Publication Details	Algorithm/Technique Used	Advantages	Limitations
1.	Nakamoto, Satoshi. "Bitcoin: A Peer-to-Peer Electronic Cash System." 2008.	Blockchain-based verification	Provides tamper-proof records and decentralized trust	High energy consumption for blockchain maintenance
2.	Rivest, R., Shamir, A., & Adleman, L. "A method for obtaining digital signatures and public-key cryptosystems." 1978.	Public Key Infrastructure (PKI)	Ensures document authenticity and integrity	Requires robust key management policies
3.	MIT Digital Diploma Initiative	Blockchain-secured digital credentials	Instant verification and reduced fraud	Adoption challenges among institutions
4.	Digitary: Secure Academic Document Exchange Platform	Encrypted cloud-based storage	Enhances accessibility and document longevity	Privacy risks if improperly managed

System Architecture



Technical Requirements Specifications

Software Requirement Specifications (SRS)

- **Coding Language**: Python 3.8
- **Libraries** : tkinter, pillow,crypography,openpyxl
- **Cryptographic Libraries** : openSSL
- **Vs Code**

Hardware Requirement Specifications

- Processor : i3 or Above.
- Hard Disk : 40 GB.
- Ram : 4 GB.

Thank you

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